$$P(x) = \begin{cases} Cx^2, & 1 < x < 4 \\ 0 & x < 1 \text{ (or) } x > 4 \end{cases}$$

① Normalising density:

$$\int_1^4 Cx^2 dx = 21C = 1 \longrightarrow C = \frac{1}{21}$$

$$P(2 \le x \le 3) = \frac{1}{21}\left(\frac{3^3 - 2^3}{3}\right) = \frac{19}{63} = 0.302$$

CDF: $F(x) = \dfrac{x^3 - 1}{63}$.